

Causal Inference and Machine Learning Analysis of Gender Quotas, Democracy, and Female Representation in Diplomatic Imagery

Ipek Gezer

Sabanci University

Istanbul, Turkey

Email: ipek.gezer@sabanciuniv.edu

Abstract—This report investigates the relationship between gender quota policies, democratic institutions, and female representation in diplomacy-related imagery. Using an image-based country-year panel merged with V-Dem indicators, the effect of strong gender quotas on female representation is estimated using a set of causal estimators including a naive mean difference, propensity-score based methods (matching and inverse propensity weighting), potential outcome modeling, and a regression-adjusted specification with uncertainty estimates. Across specifications, no statistically significant average treatment effect is found.

As an extension, supervised machine learning models are applied to evaluate whether institutional indicators can predict female representation. Country-level regression models provide modest predictive power, while a single-feature threshold-based classification approach using a democracy indicator achieves moderate discrimination. Overall, predictive performance remains limited, suggesting that institutional indicators alone are insufficient to explain cross-national variation in visual gender representation in diplomatic media.

I. DATA SOURCES

The diplomatic news and imagery data used in this study are publicly provided by the VRL Lab [1] originated from the GlobalDiplomacyNet project [2]. The Diplomatic Text Dataset hosted on Harvard Dataverse is used [3]. Democracy indicators are taken from the V-Dem Country-Year dataset (Full+Others, v15) [4].

II. DATA AND VARIABLE CONSTRUCTION

A. Outcome Variable: Female Representation in Diplomatic Imagery

For each image, the total number of detected individuals is computed as the sum of detected males and females. Images with zero detected individuals are excluded, as the female representation ratio is undefined in the absence of detected people.

Detected counts are aggregated to the country-year level, and the outcome variable is defined as:

$$female_ratio_{c,t} = \frac{female_count_{c,t}}{male_count_{c,t} + female_count_{c,t}}. \quad (1)$$

This variable captures the proportion of women appearing in diplomacy-related imagery for a given country and year and serves as the primary outcome in the analyses.

B. V-Dem Institutional Indicators

To characterize country-level institutional and democratic context, the constructed country-year outcome is merged with the V-Dem Country-Year dataset (v15). From V-Dem, a set of democracy- and gender-related indicators (e.g., electoral democracy, liberal democracy, civil liberties, gender equality, and women’s legal rights) is extracted. Importantly, the gender quota strength indicator `v2lgqugen` is also obtained from V-Dem and is central to the causal analysis.

The quota indicator `v2lgqugen` takes discrete values in $\{0, 1, 2, 3, 4\}$, reflecting increasing levels of quota strength. While multiple V-Dem indicators are used as predictive features in the machine learning component, `v2lgqugen` is used to define the treatment in the causal analysis.

III. CAUSAL INFERENCE ANALYSIS

The causal inference analysis is conducted in order to isolate the average treatment effect of strong gender quotas on female representation in diplomatic imagery, under the potential outcomes framework.

A. Research Question and Treatment Definition

The causal question of interest is: *Do strong gender quota policies increase female representation in diplomacy-related imagery?*

The V-Dem gender quota indicator `v2lgqugen`, which takes discrete values in $\{0, 1, 2, 3, 4\}$, is used. To obtain a clear and interpretable policy contrast, a binary treatment is defined at the country-year level:

$$T_{c,t} = \begin{cases} 1 & \text{if } v2lgqugen_{c,t} \in \{3, 4\} \quad (\text{High quota}) \\ 0 & \text{if } v2lgqugen_{c,t} = 0 \quad (\text{No quota}) \end{cases} \quad (2)$$

Observations with intermediate quota levels ($v2lgqugen \in \{1, 2\}$) are excluded. These categories are relatively sparse in the merged sample and yield unstable estimates when compared to either extreme. Focusing on no quota versus strong quota regimes yields a more robust and policy-relevant comparison.

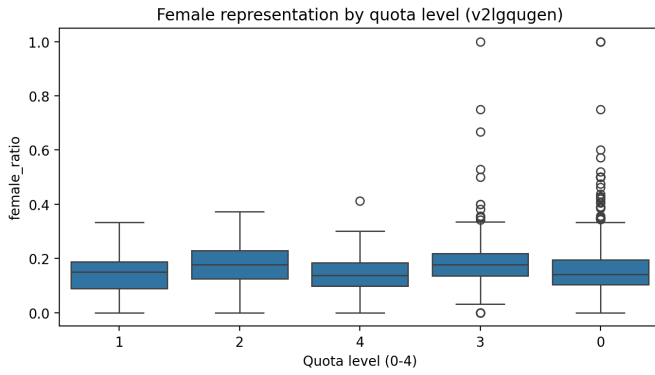


Fig. 1. Female representation by quota level (v2lgqugen 0–4).

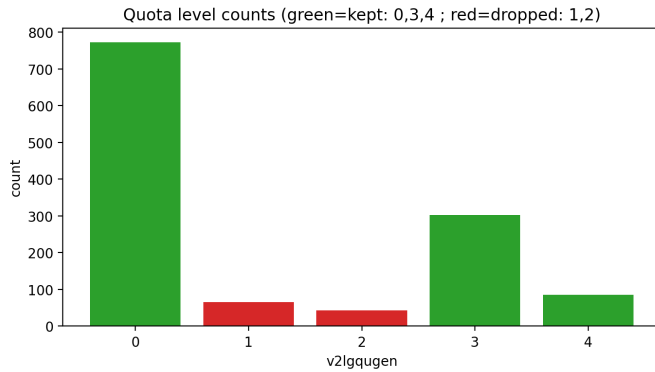


Fig. 2. Counts of quota levels. Green bars are retained (0, 3, 4) and red bars are excluded (1, 2).

B. Exploratory Evidence

Before estimating causal effects, the raw association between gender quotas and female representation is examined.

Figure 1 shows the distribution of *female_ratio* across all quota levels (0–4). The distributions largely overlap, suggesting no strong monotonic relationship in the raw data.

Figure 2 displays the number of observations per quota level. Intermediate categories (1–2) have limited support, motivating their exclusion from the causal comparison.

Finally, Figure 3 compares female representation for quota levels 0, 3, and 4 separately. This visualization checks whether levels 3 and 4 exhibit similar behavior, justifying their aggregation into a single “High quota” category.

C. Estimation Strategy

The average treatment effect (ATE) of strong gender quotas on female representation is estimated using multiple complementary approaches.

First, the naive mean difference in outcomes between treated and untreated observations is reported. Next, propensity scores are estimated using a logistic regression model with the following covariates (where available): year, liberal democracy, civil liberties, egalitarian democracy, suffrage, women’s legal rights,

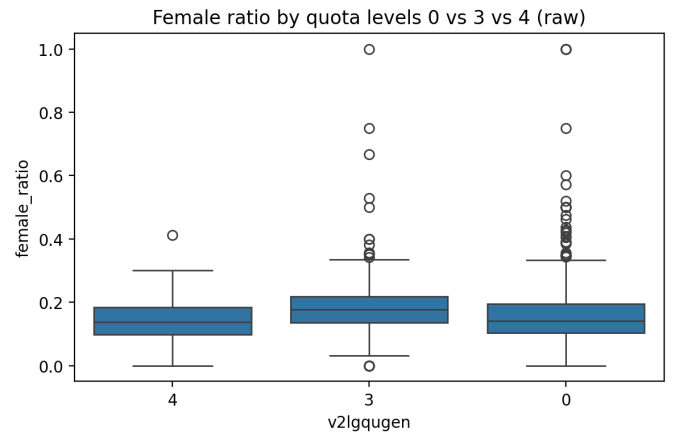


Fig. 3. Female ratio by quota levels 0, 3, and 4.

and polyarchy (V-Dem indicators). Overlap is then assessed by comparing propensity-score distributions across treatment groups.

Using the estimated propensity scores, ATE estimates are computed via (i) 1-nearest-neighbor propensity-score matching, and (ii) inverse propensity weighting (IPW) with unstabilized weights. As an additional adjusted approach, a potential outcome model (POM) is estimated by fitting separate outcome regressions for treated and control groups and averaging predicted counterfactual outcomes to obtain an ATE.

Finally, a regression-adjusted specification using OLS is reported:

$$Y_{c,t} = \beta_0 + \tau T_{c,t} + \mathbf{X}'_{c,t} \boldsymbol{\beta} + \varepsilon_{c,t}, \quad (3)$$

where $\mathbf{X}_{c,t}$ includes the same covariates used in the propensity-score model. Uncertainty is reported using cluster-robust standard errors at the country level when a country identifier is available.

D. Results

The analysis uses quota levels 0 versus 3–4 (dropping 1–2), yielding 1161 country–year observations (773 untreated, 388 treated). The propensity-score model is estimated on 1154 observations after dropping missing covariates.

Table I summarizes effect estimates across methods. While the naive mean difference is positive, adjusted methods (matching, POM, and regression adjustment) yield small negative estimates. The regression-adjusted estimate is not statistically significant.

TABLE I
CAUSAL EFFECT ESTIMATES OF STRONG QUOTAS ON FEMALE REPRESENTATION.

Method	ATE / Coefficient
Naive (raw mean diff)	0.014298
PS Matching (1-NN)	-0.007077
IPW	0.010349
POM	-0.008843
OLS (regression adjustment)	-0.006461

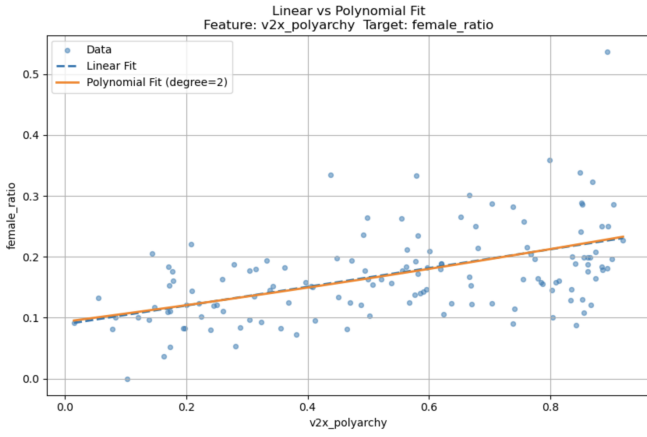


Fig. 4. Country-level female representation versus `v2x_polyarchy`. Linear and polynomial regression fits using a single predictor.

For the regression-adjusted specification, the estimated treatment effect is:

$$\hat{\tau} = -0.00646, \quad SE = 0.01307, \quad p = 0.621,$$

with a 95% confidence interval of $[-0.0321, 0.0192]$ (cluster-robust by country). Overall, the estimated effects are small and statistically insignificant, indicating no evidence that strong gender quota policies are associated with higher female representation in diplomatic imagery in this dataset.

IV. MACHINE LEARNING ANALYSIS

In addition to the causal analysis, a supervised machine learning (ML) component is implemented to assess the predictive relationship between institutional indicators and female representation in diplomatic imagery. Unlike the causal analysis, the goal of this section is purely predictive.

All ML analyses are conducted at the **country level**. Following the implementation, the data are first restricted to years ≥ 2000 and female representation and selected V-Dem indicators are then aggregated by taking country-level means, yielding one observation per country.

A. Single-Feature Regression: Linear and Polynomial Models

A simple regression setup is first considered using a single institutional indicator, `v2x_polyarchy`, as the predictor. This step serves as an initial diagnostic to assess whether a basic linear or mildly non-linear relationship exists between overall democratic quality and female representation in diplomatic imagery.

A linear regression model is estimated, followed by a second-degree polynomial regression using the same predictor. Figure 4 visualizes the fitted relationships alongside the observed country-level data.

Both models exhibit very limited explanatory power in held-out evaluation. In fact, the degree-2 polynomial specification does not improve test performance over the linear baseline (both yield negative R^2 values), suggesting that a single

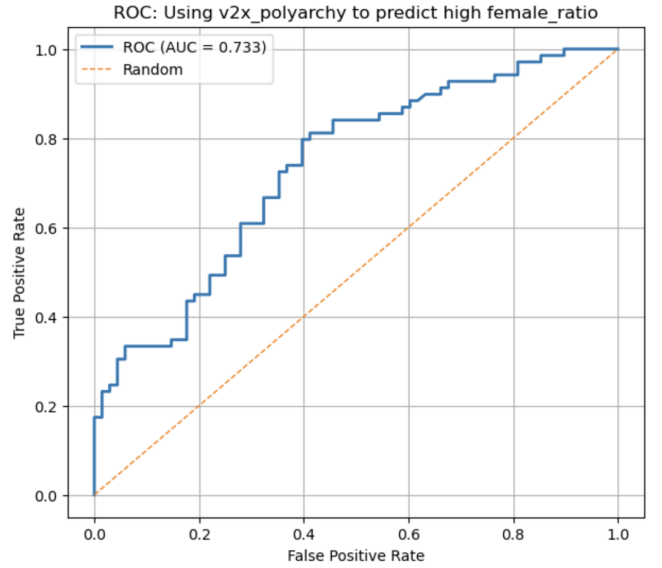


Fig. 5. ROC curve for country-level classification using `v2x_polyarchy` as the predictor.

democracy indicator is insufficient for predicting cross-country variation in female representation.

B. Binary Classification with a Single Institutional Indicator

To complement regression on continuous outcomes, a binary prediction task is evaluated at the country level. Following the implementation, countries are labeled as having high versus low female representation via a median split of the aggregated outcome `female_ratio` (threshold ≈ 0.1606). Using `v2x_polyarchy` as the sole score variable, an 80/20 stratified train-test split is performed.

Rather than fitting a parametric classifier, a thresholding rule is applied to `v2x_polyarchy`. The decision threshold is selected on the training split to maximize the F1 score, and the resulting threshold is then evaluated on the held-out test split.

The test-set performance is:

- Accuracy: 0.75
- Precision: 0.684
- Recall: 0.929
- F1-score: 0.788

The corresponding confusion matrix is:

$$\begin{bmatrix} 8 & 6 \\ 1 & 13 \end{bmatrix}$$

Figure 5 reports the ROC curve using `v2x_polyarchy` as a continuous score for predicting the high-female-representation label, yielding an AUC of 0.733. Overall, these results indicate that democratic quality contains some predictive signal, but remains insufficient on its own for highly accurate classification.

C. Multi-Feature Machine Learning Models

Linear Regression, Ridge Regression (with 5-fold cross-validated α over a log-spaced grid), and kNN regression models are evaluated using a pipeline with median imputation and standardization. Model selection for kNN is performed via 5-fold cross-validation over a candidate set of k values, and final metrics are reported on an 80/20 train–test split across countries.

Finally, the analysis is extended by incorporating a broader set of V-Dem institutional indicators as input features, including `v2x_polyarchy`, `v2x_libdem`, `v2x_civlib`, `v2x_egaldem`, `v2fsuffrage`, and `v2lgfemleg`. This multivariate setup allows assessment of whether combining multiple institutional dimensions improves predictive performance.

Linear regression, Ridge regression, and k -nearest neighbors (kNN) regression models are evaluated using an 80/20 train–test split over countries. Table II summarizes predictive performance.

TABLE II
COUNTRY-LEVEL REGRESSION PERFORMANCE WITH MULTIPLE V-DEM FEATURES (TEST SET).

Model	MSE	RMSE	R^2
Linear	0.002017	0.044914	0.201615
RidgeCV ($\alpha = 109.854$)	0.001908	0.043677	0.244969
kNN (uniform), $k = 21$	0.001914	0.043749	0.242495
kNN (distance), $k = 41$	0.001974	0.044428	0.218767

Figure 6 visualizes test-set predictions versus ground truth for the RidgeCV model. While predictions broadly follow the overall trend, substantial dispersion remains, consistent with the modest R^2 values reported in Table II.

Compared to the single-feature baselines, incorporating multiple institutional indicators improves predictive performance modestly. Ridge regression with cross-validated regularization achieves the best overall performance (lowest MSE/RMSE and highest R^2), suggesting that regularization helps stabilize estimation in the presence of correlated institutional features. kNN models perform similarly but do not consistently outperform RidgeCV in this setting, indicating that any non-linear structure captured by kNN is limited at the country-aggregate level.

Given the small number of country-level observations, reported results should be interpreted as descriptive rather than definitive.

D. Interpretation

Across all specifications, ML results indicate that institutional and democracy-related indicators contain limited but non-negligible predictive signal for female representation in diplomatic imagery. Non-linear models and classification formulations improve performance relative to simple linear baselines, yet a substantial share of variation remains unexplained. This suggests that factors beyond formal institutions—such as

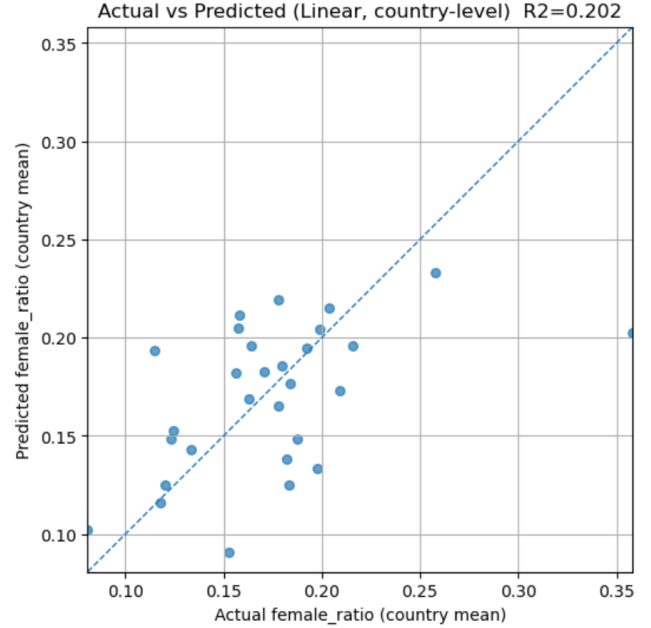


Fig. 6. Test-set predicted versus true female representation for the RidgeCV model. The dashed line indicates the 45-degree reference corresponding to perfect prediction.

media practices, cultural norms, and event selection—play an important role in shaping visual gender representation.

V. LIMITATIONS

- **Limited sample size at the country level:** The ML evaluation is conducted on country-level aggregates, which substantially reduces the number of observations.
- **Unobserved confounding (causal analysis):** The causal interpretation relies on the assumption that the included covariates capture the main confounders jointly related to quota policy and female representation. Unmeasured factors (e.g., region, economic development, media ecosystem, diplomatic engagement intensity) may still bias estimated effects.
- **Omitted cultural and societal factors:** Institutional indicators may not capture relevant cultural determinants of gender visibility in media, such as religiosity, traditional gender norms, or broader societal attitudes toward women in public life. If such factors influence both democratic institutions and media representation, they could affect both causal estimates and ML performance.

VI. CONCLUSION

This study finds no robust causal effect of strong gender quotas on female representation in diplomatic imagery. Machine learning models provide moderate predictive performance, particularly in classification settings, but overall explanatory power remains limited. These findings highlight the complexity of visual gender representation and suggest that

institutional democracy indicators alone cannot fully account for observed patterns.

VII. FUTURE WORK

Future work may expand the feature set to include proxies for cultural and societal norms. In particular, constructing a country-level “traditional vs. modern” profile (e.g., via secularization proxies, gender-norm indices, or survey-based measures) could help test whether such factors are associated with female representation in diplomatic imagery and whether they interact with institutional democracy indicators.

REFERENCES

- [1] “Vrl lab,” Website, 2024. [Online]. Available: <https://varollab.com>
- [2] “Globaldiplomacynet,” Project website, 2024. [Online]. Available: <https://www.globaldiplomacy.net>
- [3] “Diplomatic text dataset,” Harvard Dataverse, 2024, globalDiplomacyNet project dataset. Persistent ID: doi:10.7910/DVN/HYJDE0. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/HYJDE0>
- [4] “V-dem dataset (country-year: Full+others, v15),” Varieties of Democracy (V-Dem) Institute, 2025, accessed 2026-01-07. [Online]. Available: <https://www.v-dem.net/data/the-v-dem-dataset/country-year-v-dem-fullothers-v15/>